Lesson Planning

Complete Lesson Plan Book

Class: Class 7

Subject: Computer Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Introduction to Computers

Learning Objectives

• Define a computer and state its characteristics.

• Identify the IPO cycle (input, process, output).

• Classify computers by size and purpose.

• Explain the importance of computers in daily life.

Key Concepts

• A computer is an electronic device that processes data.

• Characteristics: speed, accuracy, diligence, versatility and storage.

• IPO cycle: input → process → output.

• Types: supercomputer, mainframe, minicomputer, microcomputer.

• Computers are used in schools, banks, hospitals and offices.

Teaching Methodology (Step by Step)

1. Ask students where they have seen computers being used.
2. Define a computer and write the IPO cycle on the board.
3. Explain the five characteristics with examples.
4. Classify computer types from biggest to smallest.
5. Students draw and label the IPO cycle in notebooks.

Classroom Activities

• Identify IPO stages for a washing machine and a calculator.

• Group chart: uses of computers in five fields.

Assessment Questions

1. Define a computer and write the IPO cycle.
2. Name four characteristics of a computer.
3. Give two examples each of input and output devices.

Homework

• List ten uses of computers in daily life.

Time Allocation

Discussion 5 min, teaching 20 min, drawing 10 min, quiz 5 min.

DM

Unit 2: Hardware and Software

Learning Objectives

• Distinguish between hardware and software.

• Identify input, output and storage devices.

• Define system software and application software.

• Give examples of common software.

Key Concepts

• Hardware is the physical part of a computer; software is the set of instructions.

• Input devices: keyboard, mouse, scanner; output devices: monitor, printer, speaker.

• The CPU is the brain of the computer.

• System software (like Windows) manages hardware; application software (like Word) does tasks.

• Storage devices: hard disk, USB drive, memory card.

Teaching Methodology (Step by Step)

1. Hold up a mouse and ask: is this hardware or software?
2. Sort classroom devices into input, output and storage groups.
3. Open the system unit picture and point out the CPU.
4. Explain system vs application software with examples.
5. Matching activity: device cards to category charts.

Classroom Activities

• Label a computer diagram in the lab.

• Software hunt: list all software installed on the lab computer.

Assessment Questions

1. Differentiate between hardware and software.
2. Classify: keyboard, printer, USB drive, speaker.
3. Give two examples of application software.

Homework

• Draw a computer system and label its hardware parts.

Time Allocation

Warm-up 5 min, teaching 20 min, activity 10 min, review 5 min.

Unit 3: MS Word: Creating Documents

Learning Objectives

• Open MS Word and identify its main parts.

• Type, save and open a document.

• Format text: bold, italic, underline, font size and colour.

• Align text and insert bullet lists.

Key Concepts

• MS Word is a word-processing program.

• The ribbon contains tabs like Home and Insert.

• Formatting makes documents attractive and readable.

• Documents must be saved with a proper name and location.

• Keyboard shortcuts like Ctrl+S save time.

Teaching Methodology (Step by Step)

1. Demonstrate opening MS Word from the Start menu.
2. Point out the title bar, ribbon, ruler and status bar.
3. Type a paragraph together and save it step by step.
4. Demonstrate formatting: bold, italic, font size, colour.
5. Students format their own paragraph in the lab.

Classroom Activities

• Type and format a paragraph about 'My School'.

• Shortcut drill: Ctrl+B, Ctrl+I, Ctrl+U, Ctrl+S.

Assessment Questions

1. Name four parts of the MS Word window.
2. How do you save a document? Write the steps.
3. What is the shortcut for bold and for save?

Homework

• Type a letter to your friend in MS Word and format it.

Time Allocation

Demo 10 min, guided practice 20 min, independent task 10 min.

Unit 4: MS Word: Tables and Pictures

Learning Objectives

• Insert a table in a document.

• Add and delete rows and columns.

• Insert pictures and clip art.

• Create an attractive class timetable.

Key Concepts

• Tables organise information in rows and columns.

• The intersection of a row and column is called a cell.

• Pictures make documents attractive and clear.

• Tables can be formatted with borders and shading.

• The Insert tab holds tables, pictures and shapes.

Teaching Methodology (Step by Step)

1. Show a printed timetable and ask how it is organised.
2. Demonstrate inserting a table from the Insert tab.
3. Show how to add/delete rows and columns.
4. Insert a picture and resize it.
5. Students build their own class timetable in Word.

Classroom Activities

• Create a personal timetable in MS Word.

• Design a birthday invitation with a picture and a table.

Assessment Questions

1. Write the steps to insert a 4x3 table.
2. What is a cell in a table?
3. How do you resize a picture in Word?

Homework

• Make a marksheet table of five subjects in MS Word.

Time Allocation

Demo 10 min, practice 20 min, creative task 10 min.

Unit 5: Internet Basics

Learning Objectives

• Define the internet and state its uses.

• Explain what a web browser is with examples.

• Search for information using a search engine.

• State the rules of safe internet use.

Key Concepts

• The internet is a global network connecting millions of computers.

• A web browser (Chrome, Edge) is used to open websites.

• A search engine (Google) finds information on the web.

• A website address is called a URL.

• Never share personal information with strangers online.

Teaching Methodology (Step by Step)

1. Ask students what they use the internet for.
2. Explain the internet as a network of connected computers.
3. Demonstrate opening a browser and visiting a website.
4. Perform a live search for 'national animal of Pakistan'.
5. Teach five internet safety rules.

Classroom Activities

• Guided search: find the capitals of five countries.

• Safety poster: 'Smart Internet Rules'.

Assessment Questions

1. Define the internet.
2. Name two web browsers and two search engines.
3. Write three safety rules for using the internet.

Homework

• Search the internet (with parents) and write five facts about the Markhor.

Time Allocation

Discussion 5 min, demo 15 min, guided search 15 min, safety 5 min.

Unit 6: Email: Sending and Receiving Messages

Learning Objectives

• Define email and state its advantages.

• Identify the parts of an email address.

• Compose and send an email.

• Open, reply to and forward emails.

Key Concepts

• Email sends messages electronically in seconds.

• An email address looks like name@example.com.

• The subject line tells what the email is about.

• CC sends a copy to others; attachments carry files.

• Always check the address before pressing send.

Teaching Methodology (Step by Step)

1. Compare a postal letter with an email for speed and cost.
2. Explain the parts of an email address.
3. Demonstrate composing an email with subject and body.
4. Show how to attach a file.
5. Show inbox: opening, replying and forwarding.

Classroom Activities

• Send a practice email to the teacher's address.

• Write the parts of an email address on cards and label them.

Assessment Questions

1. Write the parts of an email address.
2. What is the purpose of the subject line?
3. Write the steps to attach a file to an email.

Homework

• Write a formal email to your teacher requesting leave.

Time Allocation

Comparison 5 min, demo 20 min, practice 10 min, review 5 min.

Unit 7: MS Excel: Introduction to Spreadsheets

Learning Objectives

• Define a spreadsheet and state its uses.

• Identify rows, columns, cells and the active cell.

• Enter and format data in a worksheet.

• Perform simple calculations with formulas.

Key Concepts

• MS Excel organises data in rows and columns.

• A cell is named by its column letter and row number, e.g. B4.

• Formulas start with = and calculate automatically.

• =SUM(A1:A5) adds a range of cells.

• Spreadsheets are used for marks, budgets and records.

Teaching Methodology (Step by Step)

1. Show a paper marksheet and its Excel version.
2. Point out rows, columns, cells and the name box.
3. Students enter their five subject marks.
4. Demonstrate =SUM to total the marks.
5. Demonstrate =AVERAGE and simple formatting.

Classroom Activities

• Build a personal marksheet with total and average.

• Format the marksheet with borders and colours.

Assessment Questions

1. What is the address of the cell in column C, row 7?
2. Write the formula to add cells A1 to A10.
3. Name two uses of spreadsheets.

Homework

• Make a monthly expense sheet of your family in Excel.

Time Allocation

Demo 10 min, data entry 15 min, formulas 10 min, formatting 5 min.

Unit 8: Computer Safety and Ethics

Learning Objectives

• Define computer ethics.

• List the do's and don'ts of computer use.

• Explain the dangers of viruses and how to avoid them.

• Practise correct sitting posture at the computer.

Key Concepts

• Computer ethics are the moral rules of using computers.

• Do not copy others' work; respect copyright.

• A virus is harmful software that damages data.

• Antivirus software protects the computer.

• Sit straight, keep the screen at eye level and take breaks.

Teaching Methodology (Step by Step)

1. Discuss: is it right to copy a friend's project from the internet?
2. List computer do's and don'ts on the board.
3. Explain viruses with the 'sick computer' analogy.
4. Demonstrate running an antivirus scan.
5. Practise correct posture at the lab computers.

Classroom Activities

• Design an 'Ethical Computer User' badge.

• Checklist audit: is our lab following safety rules?

Assessment Questions

1. Write four rules of computer ethics.
2. How can you protect a computer from viruses?
3. Describe the correct sitting posture.

Homework

• Write five safety rules and paste them near your home computer.

Time Allocation

Discussion 10 min, teaching 20 min, activity 5 min, review 5 min.